National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task: Create and Containerize a C++ Application

Part 1: Develop the C++ Application

Create a Simple C++ Application:
Develop a C++ program that simulates a **Grade Calculator** for a student. The program should:

1. Accept the marks for five subjects as input.
2. Calculate the total marks, percentage, and grade based on the following criteria:
    - **90% and above:** Grade A
    - **80% - 89%:** Grade B
    - **70% - 79%:** Grade C
    - **60% - 69%:** Grade D
    - **Below 60%:** Grade F
3. Display the total marks, percentage, and grade as output.

Example of Input/Output:

- `Enter marks for subject 1: 85`
- `Enter marks for subject 2: 90`
- `Enter marks for subject 3: 78`
- `Enter marks for subject 4: 88`
- `Enter marks for subject 5: 92`
- `Total Marks: 433`
- `Percentage: 86.6%`
- `Grade: B`

Part 2: Dockerize the C++ Application

Create a Dockerfile:

In the same directory as your C++ source file, create a Dockerfile to containerize the application. The Dockerfile should:

1. Use an appropriate base image with C++ build tools installed (e.g., `gcc` or `clang`).
2. Copy the C++ source file into the container.
3. Compile the C++ code within the container.
4. Set the default command to execute the grade calculator program.

Build the Docker Image:
Use the Docker CLI to build an image from the Dockerfile. Name the image appropriately (e.g., `yourusername/grade-calculator`).

Test the Image Locally:
Run the container locally to verify that the Grade Calculator works correctly within the container.

Part 3: Push the Docker Image to Docker Hub

Login to Docker Hub:
Log in to your Docker Hub account using the command line.

Tag the Docker Image:
Tag the built Docker image in the format required for your Docker Hub repository.

Push the Docker Image to Docker Hub:
Push the tagged image to your Docker Hub account.

Part 4: Run the Container from Docker Hub

Pull the Image from Docker Hub:
On a different machine or a clean environment, pull the Docker image using the `docker pull` command.

Run the Container:
Run the container using the pulled image and test the functionality of the Grade Calculator within the container.

Deliverables:

1. **C++ application code** (`grade_calculator.cpp`).
2. **Dockerfile** used to build the container.
3. **Docker image** pushed to Docker Hub (provide the link to the image).
4. **Screenshots or output** showing the container running successfully.

THE END
